

Brac University
Department of Computer Science and Engineering
CSE423: Computer Graphics
Assignment 02

1. Given a line segment from $(-5, 2)$ to $(25, 40)$, construct the parametric equation $P(t)$ of the line. Using the parametric equation, determine the coordinates of the point where $t = 2/3$. Also, check whether the point corresponding to $t = 5$ lies inside the line segment. [2]
 - b) What is the role of the parametric equation in the Cyrus-Beck algorithm? [2]
 - c) A viewing window from $(-15, 5)$ to $(60, 120)$ is given. Using the Cyrus-Beck algorithm, determine whether the line segment from $(20, 30)$ to $(90, 80)$ is fully accepted, rejected, or needs to be clipped. If it needs to be clipped, find out the clipped line's endpoints. [6]
2. a) In which scenarios does the Cohen-Sutherland Algorithm fail? What are the drawbacks of this algorithm? [4]
 - b) A clip region from $(-40, -20)$ to $(20, 30)$ is given. Check whether the line segment $(-10, -50)$ to $(15, 25)$ is fully accepted, rejected, or partially inside using the Cohen-Sutherland Algorithm. If it's partially inside, run the algorithm to calculate new endpoints for the line segment that is inside the viewing window. [6]